

# Pierpasquale Colagrande

Computer Vision Data Scientist

Stockholm, Sweden

## Experience

### Computer Vision Data Scientist

Arkion • Full-time

Mar 2026 - Present  
Stockholm, Sweden

R&D in computer vision to improve the company's electrical power grid component analysis platform

- Implemented state-of-the-art CNN and transformer-based vision models and applied advanced deep learning techniques with PyTorch
- Developed and deployed a synthetic data generation pipeline using Blender 3D modeling and Modal
- Maintained and developed an internal CLI tool for quick deployment and update of machine learning models
- Created, cleaned and maintained datasets, trained, tested and deployed models for image classification and object detection
- Achieved an average of ~90% accuracy (mAP 50-95) on multiple object detection models
- Implemented analytics SQL queries for datasets and machine learning models by means of Google BigQuery and Metabase

Additional initiatives

- Implemented and deployed an autoresearch AI agent for improving generation of synthetic data
- Fine-tuned prompts for object detection tasks using VLMs
- Developed agent skills and MCP tools for internal purposes
- Implemented DevOps/MLOps pipelines with GitHub, Docker, AWS and Terraform
- Instructed annotation teams on how to build datasets for image classification and object detection tasks
- Designed and conducted technical interviews and assignments for data science roles across multiple seniority levels

### Artificial Intelligence Engineer

Blimp.ai • Full-time

Feb 2024 - Feb 2026  
Milan, Italy

R&D for the company's flagship product, a computer vision sensor designed for urban monitoring and analysis

- Increased inference speed 32x, from 3 to 97 FPS by optimizing the codebase and porting it to NVIDIA Jetson Orin Nano with DeepStream
- Achieved a 63% increase in people-counting accuracy by upgrading to Ultralytics YOLOv11-Small + BoT-SORT and optimizing hyperparameters
- Achieved 92% accuracy on staff member classification by training image classification networks
- Created, cleaned, and prepared datasets for image classification and object detection
- Reduced offline sensors by 65% by designing and developing a robust input module for reliable IP camera connectivity using GStreamer
- Enabled a single sensor to handle multiple cameras without FPS loss, leveraging NVIDIA DeepStream optimizations

Development of additional features

- Created and maintained an updater tool for seamless deployment of packages, patches and new features to sensors in production
- Created and maintained a GUI-based tool for remote sensor management
- Developed REST APIs to deliver features tailored to clients' needs

### Data Scientist

Lutech • Full-time

May 2023 - Jan 2024  
Cinisello Balsamo, Italy

AI R&D projects

- Fine-tuned vision models (CNNs, ViTs) for detection and classification on MRI images & multimodal datasets, using Hugging Face
- Designed and developed an intelligent hydraulic plant automation system leveraging Reinforcement Learning and Physics-Informed Deep Learning
- Created algorithms to convert hydraulic plant schematics into Deep Neural Network (DNN) architectures via modular building blocks

AI consulting projects

- Delivered Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) solutions, including chatbots and text document analysis
- Implemented vision systems with Vertex AI and Google Cloud Platform
- Built CI/CD pipelines using Docker, Kubernetes, and Mia-Platform

### Master's Thesis Intern

Univrses • Internship

Mar 2022 - Jul 2022  
Stockholm, Sweden

R&D of a copy-paste data augmentation method for generating synthetic object detection datasets, deployed in the company's training pipeline

## Education

### Artificial Intelligence

University of Bologna • Master's Degree

Sep 2019 - Feb 2023  
Bologna, Italy

Topics: Machine & Deep Learning, Natural Language Processing, Computer Vision, Reinforcement Learning, Knowledge Engineering • GPA: 3.73/4

### Computer Science

University of Bari • Bachelor's Degree

Sep 2015 - Apr 2019  
Bari, Italy

Topics: Algorithms and Data Structures, Databases, Object-Oriented Programming, Software Engineering • GPA: 3.86/4

## Publications

**CRISPRLearner: A Deep Learning-Based System to Predict CRISPR/Cas9 sgRNA On-Target Cleavage Efficiency** 

MDPI, Electronics • December 2019

## Projects

**MPRNet: CNN for Super Resolution** 

Sep - Dec 2022

**Deep Reinforcement Learning for Atari 2600** 

Jul - Sep 2022

**BERT Fine-tuning for Question Answering** 

Jan - Apr 2021

**Conditional GAN for Face Generation** 

Apr - Jul 2020

More on **GitHub** 

## Technical & Soft Skills

**Artificial Intelligence:** Image Classification, Object Detection & Tracking, Edge AI, CNNs, Transformers, VLMs, LLMs, Agents, RAG, Machine & Deep Learning, Reinforcement Learning, Knowledge Graphs

**Languages:** Python, Shell Scripting, SQL, HTML/CSS, PHP, JavaScript, C, C++, Java, Scala

**Cloud & Databases:** AWS, Modal, Google Cloud Platform, Vertex AI, BigQuery, Metabase, SLURM, Dremio

**DevOps & MLOps:** Git, GitHub, GitLab, Bitbucket, Docker, Kubernetes, GitHub Actions, Terraform, Weights & Biases, MLflow, TensorBoard

**Frameworks & Libraries:** PyTorch, TensorFlow, OpenCV, NVIDIA Jetson, GStreamer, DeepStream, TensorRT, ONNX, LangChain, crewAI, smolagents, Keras, OpenAI, OpenAI Gym, Scikit-learn, Hugging Face, Pandas, NumPy, FastAPI, pytest

**Agile:** Scrum, Kanban, Jira, Confluence, Notion

**Soft Skills:** Problem Solving, Leadership, Continuous Learning, Proactivity, Curiosity, Ambition, Teamwork, Communication, Attention to Detail